

The Augustine Problem in Agents: Chronoception Must Be Installed

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Abstract

LLM agents are trained on tokens and deployed in wall-clock time. They cannot perceive the wall-clock, and we prove this perception cannot be acquired from any token-only training loss. The Chronoception Impossibility Theorem (CIT) makes the claim precise; the Three Times ontology ($\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$) names the projections a policy must align; the Augustine Problem is that alignment failure. On ChronoBench (9 sub-capabilities $\times$ 14 agents from 7 vendors $\times$ 2 settings, $\sim 5,900$ trajectories including the first full $9 \times 2 \times 30$ panel of a reasoning model): every agent leaves $\geq 89\%$ of its wall-clock budget unspent (panel median CAR ≤ 0.104 ; non-reasoning median 0.014); date injection does not move CAR ($|\Delta \text{CAR}| \leq 0.01$). The Reverse-Scaling Theorem, a direct corollary of CIT for budget-honoured reasoning K , states $\mathbb{E}[|\rho| \mid K]$ is monotone non-decreasing in K ; confirmed four ways — intra-model (o4-mini effort ladder $1.23 \rightarrow 1.54 \rightarrow 1.67$), cross-model (Sonnet 4.6 $\pm$ thinking, sign flip $+0.07 \rightarrow -0.17$), within-model (prospective $+0.16$ vs. retrospective -0.25 on the same tasks), and cross-vendor (GLM-5.2 sign flip $+0.16 \rightarrow -0.29$, independent training pipeline). The Calibration Catastrophe follows: nominal 90% CIs achieve 0–77% coverage panel-wide. The Injection Tell: 3/3 consumer web-chat products inject the current date; 0/3 developer tools do — three closed labs converging independently. A toy positive control (Qwen2.5-1.5B + LoRA SFT on wall-clock-supported targets) crosses the Augustine threshold $\varepsilon^* = 0.20$ on T3.1 after ~ 60 s of training — the constructive converse of CIT. The Agentic Timeline Hypothesis, $T_{\text{max}} \propto 1/\varepsilon$, is quantitatively supported on METR HCAST (14,709 runs; Mann–Whitney one-sided $p = 0.040$; permutation $p = 0.037$). Code, trajectories, and locked OSF pre-registration released.

1 Introduction

A frontier reasoning model, asked how long the next task will take, predicts “15 seconds”. It thinks for ninety. It answers in two. Asked afterwards how long the task took, it reports “0.04 seconds”. Asked for a 90% confidence interval on that duration, the interval contains the truth 0% of the time. None of the four numbers is a bug or a misread prompt. The pattern is joint, panel-wide, and structurally predictable: we observe it across 14 agents from 7 vendors on $\sim 5,900$ trajectories.

The Augustine Problem. Every training loss in current practice — next-token cross-entropy, SFT, RLHF, RL with verifiable rewards, reasoning supervision — is a functional of token sequences. None contains wall-clock time in its support. Yet the same models are deployed as agents in continuous physical time, where budgets are measured in seconds and consequences unfold without regard to step count. We call this representational gap the Augustine Problem, after Augustine’s confession that he knew what time was until asked to explain it.¹ The gap is structural: we prove that under any token-only loss, the policy

¹We use chronoception in the cognitive-science sense of Wittmann (2009): perception of one’s own work duration.

gradient carries no signal aligning τ_{wall} with any internal representation. Chronoception cannot be acquired from token-only training; it must be installed.

Position. We make that claim precise, measure the failure at scale, audit the workaround the industry has converged on, and specify what closing the gap would take. Our support is layered rather than a single benchmark: (a) ChronoBench, a 9-capability diagnostic across 14 agents; (b) the Closed-Lab Injection Atlas, an audit of 11 harnesses; (c) a toy positive control demonstrating CIT’s constructive converse; (d) horizon-scaling evidence on 14,709 METR HCAST runs.

Contributions.

- The Augustine Problem as a named structural diagnosis with the Three Times ($\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$) ontology (§2).
- Two theorems. CIT (Theorem 1): no token-only loss aligns τ_{wall} with any internal representation. Reverse-Scaling (Theorem 2): under CIT with budget-honoured reasoning K , $\mathbb{E}[|\rho| \mid K]$ is monotone non-decreasing in K . Confirmed four ways (§6).
- ChronoBench: 9-cap $\times$ 14-agent $\times$ 2-setting diagnostic, $\sim 5,900$ trajectories, first full $9 \times 2 \times 30$ panel of a reasoning model (§4).
- The Injection Tell. Cross-vendor natural experiment; toy positive control crossing ε^* (§7, §6).
- Agentic Timeline deployment bound $T_{\text{max}} \propto 1/\varepsilon$ with quantitative HCAST support (Mann–Whitney $p = 0.040$) and spatial generalisation SIT (§8, App.).

2 The Three Times of an Agent

Definition 1 (Three Times). For a trajectory $\tau = (s_0, a_0, \dots, s_n)$: $\tau_{\text{wall}}(\tau) := t(s_n) - t(s_0)$ (external), $\tau_{\text{step}}(\tau) := n$ (policy invocations), $\tau_{\text{self}}(\tau) := \Pi(s_n^{\text{output}})$ (agent-reported duration).

Grounded chronoception requires $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$, where $\langle \Delta t \rangle$ is mean per-step wall-clock cost. The first approximation is L2 (step count tracks wall-clock); the second is L3 (narration tracks reality). The Augustine Problem is joint drift across all three.

Why three, not two or four? $\tau_{\text{wall}} \neq \tau_{\text{step}}$: a 10-step trajectory takes 5s on Haiku, 200s on a thinking model. $\tau_{\text{wall}} \neq \tau_{\text{self}}$: agents mis-report by up to $34\times$ (§6). $\tau_{\text{step}} \neq \tau_{\text{self}}$: an agent can produce 6 steps and report “10 seconds” without a grounded mapping. A single-time ontology loses either the training-loss argument (acts on τ_{step}), the deployment argument (acts on τ_{wall}), or the narrative argument (acts on τ_{self}); four times (e.g., tool-read wall-clock) subsume into τ_{wall} once the tool resolves. The three loci are minimal and complete.

3 The Augustine Problem and CIT

Definition 2 (Augustine Problem). Agent A exhibits the Augustine Problem on task family $\mathcal{T}$ if $\mathbb{E}_{\tau \sim \pi_A}[|\tau_{\text{wall}} - \tau_{\text{step}} \langle \Delta t \rangle| + |\tau_{\text{wall}} - \tau_{\text{self}}|] > \varepsilon_0$ for calibration tolerance ε_0 .

Assumption (token-only loss support). A training procedure is in the token-only regime if $\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}}[\ell(f_\theta(x_{<i}), x_i)]$ satisfies (i) $\mathcal{D}$ is a distribution over finite token sequences with no field carrying wall-clock duration and (ii) ℓ is a function of the model’s output token distribution alone. Next-token CE, SFT, RLHF, RL with verifiable rewards, and reasoning supervision all satisfy this.

Theorem 1 (Chronoception Impossibility, CIT). Let $\mathcal{L}$ be a token-only loss. For any strictly increasing reparametrisation $\phi : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$, $\mathcal{L}(\theta)$ is invariant under $\tau_{\text{wall}} \mapsto \phi(\tau_{\text{wall}})$ on the data manifold. Equivalently, $\nabla_\theta \mathcal{L}$ carries zero gradient signal aligning τ_{wall} with any internal representation.

Sketch. $\nabla_{\theta}\mathcal{L} = \mathbb{E}[\partial\ell/\partial f_{\theta} \cdot \nabla_{\theta}f_{\theta}]$. Neither factor depends on the wall-clock time at which the sample was generated: tokens carry no timestamps; ℓ is a function of the output distribution alone. The gradient is invariant to any reparametrisation of τ_{wall} . $\square$

Scope. CIT does not claim there is no representation of time inside the policy; it claims any such representation is not produced by the loss. Explicitly out of scope: (a) harness-installed chronoception (the Injection Tell, §7); (b) tool-output-installed chronoception (a date tool call grounds the policy through the return value, not the loss); (c) future losses with τ_{wall} as argument (Paper 2). CIT precludes the widely held inference that scale, longer reasoning, or larger context will close the gap.

Three empirical predictions. P-CIT-1: under wall-clock budgets, CAR stays bounded away from 1 regardless of capability (§5). P-CIT-2: prompt-level τ_{wall} information cannot move action-axis behaviour: $|\Delta\text{CAR}|$ is small under Setting B (§5). P-CIT-3: calibration tooling cannot close self-duration coverage because the calibration target is not in the loss support (§6). All three confirmed below.

Augustine threshold. We adopt $\varepsilon^* = 0.20$ as an operational reporting threshold. We state plainly what it is and is not. It is not derived from a measured human baseline: human duration perception is itself imperfect and scale-dependent (Wittmann, 2009; Eagleman, 2008), but that literature reports discrimination thresholds rather than the retrospective estimation error our ρ measures, and we know of no published human value directly convertible to it. $\varepsilon^* = 0.20$ is the round figure one order of magnitude below a $2\times$ misestimate ($|\rho| = 0.30$), chosen before the panel was run and fixed thereafter. Nothing in our argument depends on the constant: the closest agent sits at $\varepsilon = 0.536$, and $\text{score}_{T_2} \geq 0.92$ on every agent puts a floor of $\varepsilon \geq 0.46$ under the whole panel irrespective of the narrative axis, so the qualitative conclusion is invariant to any threshold in $[0.1, 0.45]$. A task-matched human panel would let the threshold be earned rather than stipulated; that is the highest-priority follow-up (§10).

4 ChronoBench

ChronoBench’s design philosophy is single-axis isolation: every instance loads on exactly one of $(\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}})$, so a failure is unambiguously attributable to the axis under test. Table 1 lists the 9 sub-capabilities and pairs each with the diagnostic question a deployment auditor should ask.

Table 1: The 9 sub-capabilities of ChronoBench. Each row isolates one chronoceptive failure mode.

Axis	Sub-cap	Tests	Auditor question
τ_{wall}	T1.1 Clock awareness	Today’s date / current time	“Does it know when it is?”
	T1.2 Elapsed-time	Time since conversation start	“Does it track elapsed time?”
	T1.3 Deadline tradeoff	Strategy under wall-clock deadline	“Does it re-plan under pressure?”
τ_{step}	T2.1 Step-budget	Produce exactly N steps	“Can it count its steps?”
	T2.2 Step-to-wall	Compute $N \times \Delta t = \tau$	“Can it convert step to wall-time?”
	T2.3 Wall-budget	Honour a wall-clock budget (L2)	“Does the budget change behaviour?”
τ_{self}	T3.1 Retrospective ρ	Duration reported after task	“How long did it just take?”
	T3.2 Prospective ρ	Duration predicted before task	“How long will this take?”
	T3.3 Calibration	90% CI over self-duration	“How confident in its duration?”

Panel and settings. Fourteen agent configurations from seven vendors — OpenAI (gpt-4o-mini, gpt-4o, gpt-5.1, o3, o4-mini at three effort levels); Anthropic (Haiku 4.5, Sonnet

4.6, Sonnet 4.6 + extended thinking); open-source (Qwen2.5-7B, DeepSeek-R1-Distill-Qwen-14B); and a cross-vendor Chinese-lab expansion (Z.AI’s GLM-5.2, MiniMax-M2.7, Moonshot’s Kimi-K2.6, Alibaba’s Qwen3.6-27B) via Vultr Inference. Setting A uses a vanilla prompt; Setting B prepends “Current date and time: {ISO}”. 30 trajectories per (model, capability, setting) cell, $\sim 5,900$ total, with the Sonnet-thinking variant carrying the first full $9 \times 2 \times 30$ panel coverage of any reasoning model.

5 The Action Axis: L2 Step-Clock Conflation

P-CIT-1 says the action axis cannot honour wall-clock budgets. We ask each agent “work on this task for B seconds; do not stop early”, vary $B \in \{60, 300, 900, 1800, 3600\}$ s, and measure $\text{CAR} := \tau_{\text{wall}}^*/B$.

No panel agent’s median CAR exceeds 0.104: the non-reasoning panel sits at 0.014 under Setting A and 0.016 under B, with $|\Delta\text{CAR}| \leq 0.01$ for every model. Across five non-reasoning generations $|\rho|$ falls 94% (§6) while median CAR stays in $[0.004, 0.050]$: narrative axis is text-trainable; action axis is not. Route B-1 extended T2.3 to reasoning models and Chinese labs: median $\text{CAR} \in [0.007, 0.104]$, all $\ll 1$, injection $|\Delta\text{CAR}| \leq 0.005$. The pooled medians hide the one place the failure changes shape. At $B = 60$ s every reasoning model sits an order of magnitude above its pooled median (o3 0.225, o4-mini 0.159), and Sonnet-thinking overshoots outright ($\text{CAR} = 2.22$): the extended-thinking policy cannot stop inside a minute. Under-spend and overshoot are one failure of $\tau_{\text{wall}} \approx \tau_{\text{step}} \langle \Delta t \rangle$ read at two budget scales — the sign of the miss is set by the policy’s default generation length, not by the budget.

Proposition 1 (Injection cannot install action-axis chronoception). Let A be a token-only-regime agent; A_B the same policy under Setting B. Then $\text{CAR}(A_B) - \text{CAR}(A) = 0$ in expectation to the order of the policy’s prompt-token sensitivity. Injection is a narrative-axis intervention: it can rewrite what the agent says but not what the agent does with time.

Sketch. By CIT, the policy’s prefix-to-action map is time-reparametrisation-invariant on the training manifold. Prefix-level τ_{wall} information modifies output-token generation, not the wall-clock realisation of the action sequence. Empirical $|\Delta\text{CAR}| \leq 0.01$ across the 14-cell panel is direct measurement. $\square$

L1, measured rather than assumed. Sweeping B on a fixed task is also the design L1 (Agentic Parkinson) needs, and one that T1.3 cannot supply: its prompts pair each deadline with a different question, confounding deadline with task size. Fitting the budget elasticity $e := d \log_{10} \tau_{\text{wall}}^* / d \log_{10} B$ — Parkinson’s law in full is $e = 1$, indifference is $e = 0$ — gives panel median $e = 0.089$, with 10/13 bootstrap CIs strictly above zero: a $60\times$ budget buys $1.4\times$ the work. L1 is present and directionally correct at a tenth of the required strength. We previously asserted its contrapositive; our own corpus refutes that. T2.3 is one task and its prompt instructs the agent to spend the budget, so e bounds spontaneous Parkinson from above (App.).

L2 is the load-bearing empirical claim: it anchors ε above ε^* for every panel agent and bounds the Agentic Timeline regardless of capability scaling.

6 The Narrative Axis: Closure, Ceiling, Catastrophe

Across five frontier non-reasoning generations $|\rho|$ falls 94% ($+1.12 \rightarrow +0.07$; App.). The closure is real and is bounded three ways: a theorem that reasoning-token expansion reverses it, a calibration catastrophe that persists where the surface closes, and a positive control that shows what training would actually close it.

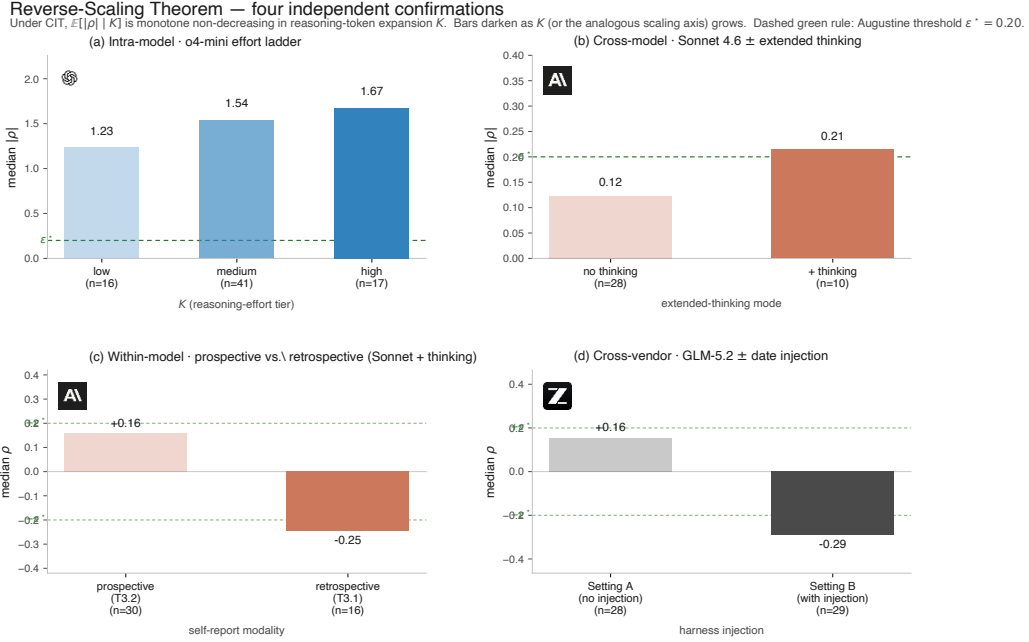


Figure 1: Four independent confirmations of Theorem 2. (a) Intra-model o4-mini effort ladder 1.23 $\rightarrow$ 1.54 $\rightarrow$ 1.67. (b) Cross-model Sonnet 4.6 $\pm$ thinking, median $|\rho| = 0.12 \rightarrow 0.21$ (Setting A, $n = 28/10$; signed median $+0.07 \rightarrow -0.17$); the sign flip is firm under Setting B (-0.25 , CI $[-0.34, -0.11]$, $n = 16$). (c) Within-model: Sonnet-thinking prospective ($+0.16$) vs. retrospective (-0.25) on the same tasks under Setting B — sign flip. (d) Cross-vendor: GLM-5.2 (Z.AI, no shared training pipeline with Anthropic), Setting A $\rightarrow$ B: $+0.16 \rightarrow -0.29$. The three Chinese-lab reasoning models (GLM-5.2, Kimi-K2.6, Qwen3.6-27B) collectively median $\rho \in [-0.97, -0.29]$ on Setting B — Hidden-Time as a class property.

6.1 The Reverse-Scaling Theorem

Assumption (budget-honoured reasoning). A reasoning-budget knob K is honoured if $\mathbb{E}[\tau_{\text{wall}} \mid K]$ is strictly increasing in K . The o4-mini reasoning_effort ladder, Anthropic extended-thinking budget, and Kimi/GLM reasoning modes satisfy this empirically; the DeepSeek-R1-Distill-14B max_completion_tokens sweep does not (App.: the model declines to spend the budget), and falls outside the assumption.

Theorem 2 (Reverse-Scaling). For a fixed agent under Assumption 3 (hence Theorem 1) with a budget-honoured knob K , $\mathbb{E}[|\rho| \mid K]$ is monotone non-decreasing in K .

Sketch. Under CIT, $p(\tau_{\text{self}})$ is invariant to τ_{wall} : no gradient signal in training ties them. Inference-time $p(\tau_{\text{self}} \mid K) = p(\tau_{\text{self}})$ for all K in the trained range. By assumption, $\tau_{\text{wall}}(K)$ is strictly increasing. Therefore $|\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|$ is monotone non-decreasing in K in expectation. The sign of ρ turns negative once hidden CoT dominates τ_{wall} , because $p(\tau_{\text{self}})$ stays anchored to the surface output; the theorem constrains only $|\rho|$, and o3 is the panel cell that grows $|\rho|$ without flipping sign (App.). $\square$

6.2 The Calibration Catastrophe

Asked to wrap self-duration in a 90% CI, every non-reasoning panel agent under-covers by ≥ 40 pp; GPT-4o reaches 0%. Standard calibration tooling (Guo et al., 2017; Kadavath et al., 2022) cannot apply because the calibration signal is not in the loss support (Theorem 1). Two cells cross the pre-registered P11 threshold of 0.5: Sonnet-thinking (77%, $n = 30$, Wilson CI $[0.59, 0.88]$) and Kimi-K2.6 (55%, $n = 29$, $[0.38, 0.71]$); o4-mini lands exactly on

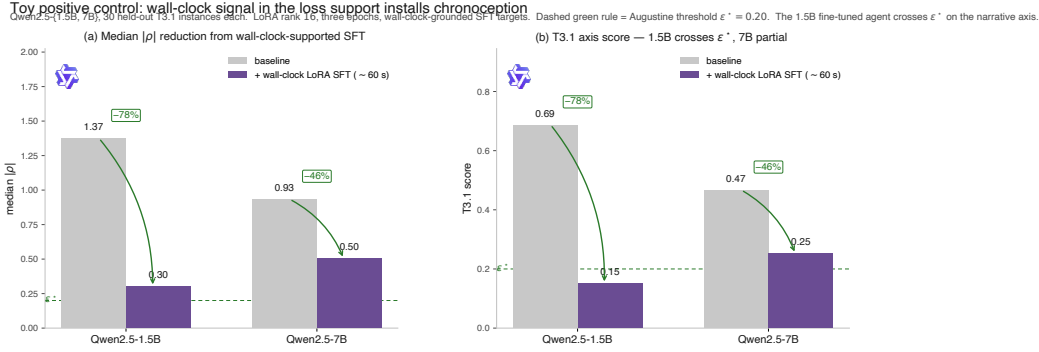


Figure 2: Toy positive control. Baseline (grey) vs. wall-clock-supported LoRA SFT (violet, ~60s of training). Left: median $|\rho|$ reduction (78% on Qwen2.5-1.5B; 46% on 7B). Right: T3.1 axis score — 1.5B fine-tuned crosses the Augustine threshold $\epsilon^* = 0.20$; 7B is partial (0.252). Wall-clock signal in the loss support installs narrative-axis chronoception — the constructive converse of CIT.

it at 15/30 and so does not cross. All three are reasoning-mode, and all three have wider stated CIs relative to their own τ_{wall} than the non-reasoning panel does. The mechanism is not learned chronoception but arithmetic: reasoning inflates τ_{wall} , $p(\tau_{\text{self}})$ stays anchored to the default vocabulary, wider CI accidentally contains the wall-clock. Both cells rest on $n = 30$, below the $n \geq 100$ bar our pre-registered F4 sets for a falsification, so we report them as threshold crossings and not as refutations — but we report them. They are CIT-consistent, and they sharpen P11’s next revision into a proper scoring rule (Brier / CRPS) rather than raw coverage.

6.3 A positive control: CIT’s constructive converse

We take Qwen2.5-1.5B and 7B, generate 500 SFT pairs of the form “{response}. This task took approximately $\{\tau_{\text{wall}}\}$ seconds.” with $\pm 15\%$ Gaussian noise on the reported duration, and LoRA fine-tune (rank 16, three epochs). The token-level CE loss is unchanged; the training targets carry wall-clock signal. On 30 held-out T3.1 instances (Fig. 2): median $|\rho|$ falls from 1.37 to 0.30 on 1.5B (78% reduction) and from 0.93 to 0.50 on 7B (46%). The 1.5B fine-tuned T3.1 score (0.151) crosses ϵ^* ; 7B partial (0.252). The sign of ρ flips from over-report to under-report at both scales. Wall-clock signal in the loss support installs narrative-axis chronoception. A full installation reaching all nine sub-capabilities requires action-axis grounding, which target-side annotation cannot supply (Paper 2).

7 The Injection Tell: A Cross-Vendor Natural Experiment

Consumer products built on these same models feel chronoceptively grounded to the people using them. They are not. Closed labs route wall-clock information into the token stream — by M1 system-prompt insertion, M2 tool calls, or M3 browser side effects — so that the harness perceives time on the model’s behalf. That three labs converged on this independently is the strongest non-experimental evidence we have that the gap is real.

Three closed labs converge on the same engineering choice for consumer products, independently, with substantial commercial incentive to ship a cheaper alternative. The within-Anthropic contrast is sharpest: the same Claude 4.6 model injects via `<knowledge_cutoff>` on Claude.ai, yet through the raw API refuses and discloses its training cutoff instead. The model is one; the chronoception is the harness’s. Setting B raises T1.1 pass to $\geq 96\%$ but leaves CAR, $|\rho|$, and coverage largely untouched — prompt information is not a policy representation.

Table 2: Injection Atlas: M1 system-prompt date injection by product tier (verbatim leaked-prompt evidence, asgeirtj corpus).

Product tier	M1 rate	Examples
Consumer web-chat	3/3 (100%)	ChatGPT, Claude.ai, Gemini all inject
Raw API	1/4 (25%)	only GPT-5.1 injects; o3, gpt-4o, Anthropic API do not
Developer tools (IDE/CLI)	0/3 (0%)	Cursor, GitHub Copilot CLI, Perplexity Computer

8 ε and the Agentic Timeline

Definition 3 (Chronoceptive Calibration Error). $\varepsilon(A)$ is the mean of the axis scores actually measured, each taken as a median over that axis’s trajectories: $\text{score}_{T_2} := \text{med } |\text{CAR} - 1|$ on T2.3, $\text{score}_{T_3} := \text{med } |\rho|$ on T3.1. An axis with no trajectories is dropped and the remaining weights renormalised — never imputed as zero. Medians rather than means because two harness artefacts (a stalled gpt-4o-mini request, $\tau_{\text{wall}} = 14.9$ h against a 900 s budget) would otherwise decide the bottom of the ranking.

No panel agent crosses $\varepsilon^* = 0.20$ (Fig. 3); the closest, Sonnet 4.6, sits at $\varepsilon = 0.536$, $2.7\times$ above. The action axis is why: $\text{score}_{T_2} \in [0.92, 1.00]$ on every agent, so even driving score_{T_3} to zero — which would require crossing both CIT and Theorem 2 — leaves $\varepsilon \geq 0.46$, still $2.3\times$ above threshold. Adding extended thinking to Sonnet 4.6 moves ε the wrong way ($0.536 \rightarrow 0.566$ under Setting A, $0.537 \rightarrow 0.585$ under B), and the o4-mini effort ladder moves it further still (ρ axis, $1.23 \rightarrow 1.54 \rightarrow 1.67$ from low to high effort). More reasoning compute moves the agent away from the threshold.

A correction to the ε convention. Our earlier draft scored an unmeasured axis as 0 while leaving its weight in the denominator. The α axis is never populated under this protocol — T2.3 is the only capability carrying a wall-clock budget and it is a τ_{step} -axis capability — so that convention credited every agent with a free perfect third, and credited an agent with a partial corpus (o4-mini, which had no T2.3 cell in the original run) with two. The ε reported here follows Definition 3 instead. The correction raises every ε , so it strengthens rather than rescues the threshold claim; it changes the ordering in one place, moving o4-mini from mid-panel to the bottom, which is where its $|\rho| = 1.54$ belongs.

Hypothesis 1 (Agentic Timeline). $T_{\text{max}}(A) \propto \frac{1}{\varepsilon(A)} \cdot f(\text{CAR}(A))$, with f monotone-increasing in CAR.

L2 does not scale with capability; Theorem 2 grows $|\rho|$ with reasoning K . Therefore T_{max} scales with neither. The autonomous-agent timeline is bounded by chronoception, not by intelligence and not by reasoning compute.

Pre-registered P12 (quantitative). Slope of success-rate decay with $\log T$ scales as $-(1 - \text{CAR}(A))$. On METR HCAST (14,709 runs after filtering, 20 frontier models), five reasoning models cluster at mean $|\text{slope}| = 0.318$ vs. 0.264 for fifteen non-reasoning models. Mann-Whitney $U = 58$, one-sided $p = 0.040$; permutation test (20,000 resamples, one-sided) $p = 0.037$; rank-biserial $r_{\text{rb}} = -0.55$ (large effect, reasoning stochastically dominates). Two-sided $p = 0.081$; we report one-sided because Theorem 2 is directional.

9 From Time to Space: Spatiotemporal Generalisation

Long-horizon benchmarks (SWE-Bench, WebArena, GAIA, MLE-Bench) require the agent to navigate space as well as time. The CIT argument goes through unchanged on any external metric the loss does not see.

Theorem 3 (Spatiotemporal Impossibility, SIT). Strengthen Assumption 3 to require $\mathcal{D}$ contains no field carrying a metric on the agent’s environment. Then $\nabla_{\theta} \mathcal{L}$ carries zero gradient signal aligning either τ_{wall} or σ_{world} with any internal representation.

Chronoceptive Calibration Error ε — 13 scored agents (7 vendors, 4 Chinese labs)

Solid bar = Setting A; hatched extension = Setting B. Dashed rule = Augustine threshold $\varepsilon^* = 0.20$; no agent crosses it, the closest sits $2.7 \times$ above.

ε = mean of the two measured axes, each a median. Red n_p marks cells below $n = 15$.

Model	ε (Setting A $\rightarrow$ Setting B)	$\varepsilon(A)$	$ CAR - 1 $	$ \rho $	n_p
★ Oracle <i>reference</i>		—	—	—	—
AI Sonnet 4.6	0.54	0.54	0.95	0.12	28
AI Sonnet 4.6 + thinking	0.59	0.57	0.92	0.21	10
Z GLM-5.2-FP8	0.62 0.64	0.62	0.99	0.25	28
gpt-5.1	0.66 0.57	0.66	0.98	0.34	11
o3 (reasoning)	0.72 0.72	0.72	0.99	0.45	21
AI Haiku 4.5	0.72 0.66	0.72	0.98	0.46	29
Kimi-K2.6	0.92 0.98	0.92	0.98	0.86	14
MiniMax-M2.7	0.92 0.66	0.92	0.99	0.85	13
Qwen3.6-27B	0.97 0.96	0.97	0.98	0.95	30
gpt-4o	1.03 0.88	1.03	1.00	1.07	30
gpt-4o-mini	1.05 1.05	1.05	0.99	1.12	30
o4-mini (reasoning)	1.26 1.25	1.26	0.99	1.54	41
Qwen2.5-7B	1.27 1.21	1.27	0.99	1.56	30

$\varepsilon^* = 0.20$

Figure 3: ε over the 13 scored agents from 7 vendors. (DeepSeek-R1-Distill-14B is the 14th panel configuration and is unscored: it declines to spend its reasoning budget, so it falls outside Assumption 6.1.) Solid bars = Setting A; hatched extension = Setting B. Dashed green rule = $\varepsilon^* = 0.20$. No agent crosses it — the closest, Sonnet 4.6, sits $2.7 \times$ above. n_p in red marks cells below $n = 15$. The Chinese-lab expansion (GLM, MiniMax, Kimi, Qwen3.6) places the framework cross-vendor.

The Agentic Timeline generalises to $T_{\max} \cdot S_{\max} \leq C/\varepsilon_{ST}(A)$ — the Agentic Frontier. Three spatial laws (SL1–SL3) mirror L1–L3; the classification of long-horizon benchmarks by (T, S) load and five future experiments E6–E10 are in Appendix.

10 Related Work, Scope, and Falsification

Concurrent work. Garikaparathi (2026) report duration self-reports on non-reasoning models (the L3 axis); we extend to reasoning models and prove Reverse-Scaling. Ma et al. (2026)’s Timely-RL installs the L1 Parkinson regime by reward shaping; on native agents we measure it present but attenuated (budget elasticity $e = 0.089$, where the law requires $e = 1$; §5). Cheng et al. (2025)’s “temporally blind” agents match L2; we add the axis split, CIT, the Atlas, and the Agentic Timeline. Gema et al. (2025) report inverse scaling in test-time compute for calibration; Theorem 2 is the structural analogue for chronoception,

with the additional content that degradation is provably monotone under CIT. Aggregate benchmarks (Jimenez et al., 2024; Zhou et al., 2024; Mialon et al., 2023; Kwa et al., 2025) correlate horizon with success; ChronoBench supplies the mechanism.

Cross-lingual replication, and a refusal asymmetry. We re-ran T1.1 and T3.1 in Mandarin Chinese at $n = 30$ per cell on three OpenAI agents. The narrative axis replicates tightly: gpt-4o-mini’s Chinese median $\rho = +1.119$ (bootstrap 95% CI $[+1.054, +1.208]$, $n = 26$) against its English baseline of $+1.12$. Setting B lifts T1.1 to 100% pass on all three models (Wilson $[0.88, 1.00]$), so the Injection Tell is language-independent.

Setting A, however, exposes a failure mode the English panel hides. In English, gpt-4o-mini refuses T1.1, citing its training cutoff. In Chinese it never refuses: 0/29 refusals and 29/29 confident date fabrications, every one of them in the training-cutoff year 2023 (modal response, translated: “Today is October 4, 2023.” — stated flatly, with no hedge). gpt-4o is intermediate (4 refusals, 4 fabrications of 8 decided). The refusal we observe in English is therefore a surface behaviour of the post-training distribution, not a representation of not-knowing: it does not survive a change of language, while the underlying absence of τ_{wall} does. This sharpens rather than softens the position. An agent that says “I do not know the date” has not thereby acquired chronoception; it has acquired an English sentence. (gpt-5.1 is the exception at 97% Setting A pass, Wilson $[0.83, 0.99]$ — stronger than its 74% English rate, and further evidence of provider-side injection on that endpoint, which we report as a confound rather than a capability.)

Scope. We do not claim current agents are useless: T1-dominated chat is harness-patchable and many production deployments inhabit that regime. We do not claim capability scaling is worthless: $|\rho|$ closes 94% across five generations. We do not claim all test-time compute fails: a system consulting a wall-clock-supported tool exits the CIT regime by construction. A contamination probe on gpt-4o and gpt-4o-mini varied the injection template across canonical/paraphrased (“Right now it is {ISO}”)/year-only/scrambled-placeholder: pass rate stayed at 100% under paraphrased, 67% under year-only (compositional); scrambled placeholder triggered correct refusal. Setting B pass reflects date-reading, not memorised keyword patterns.

Falsification. Four pre-registered forms (Anonymous Authors, 2026). (F1) A token-only agent crossing ε^* under Setting A falsifies CIT. (F2) A reasoning-tuned agent with monotone decrease in $|\rho|$ across a ≥ 3 -level, ≥ 1 -order-of-magnitude budget-honoured K sweep falsifies Theorem 2. (F3) A horizon-stratified benchmark on which reasoning shows shallower decay than non-reasoning at matched short-horizon success falsifies Hypothesis 1. (F4) A CIT-regime foundation model with $> 50\%$ T3.3 coverage without harness injection on $n \geq 100$ falsifies P11. We commit to reporting any we observe.

11 Conclusion

LLM agents are trained on tokens and deployed in wall-clock time. The Chronoception Impossibility Theorem proves this gap is structural: no token-only loss carries gradient signal aligning wall-clock with any internal representation. ChronoBench (9 sub-capabilities $\times 14$ agents $\times 2$ settings, $\sim 5,900$ trajectories) quantifies the consequence: no agent spends more than a tenth of its wall-clock budget at panel median, and the Reverse-Scaling Theorem predicts—and four independent tests confirm—that more reasoning compute makes duration confabulation worse, not better. The Calibration Catastrophe (0–77% coverage on nominal 90% CIs) and the Injection Tell (3/3 consumer products inject dates; 0/3 developer tools do) show the failure is panel-wide and industry-acknowledged.

The practical implication is sharp: chronoception bounds the autonomous-agent timeline more tightly than capability scaling does ($T_{\text{max}} \propto 1/\varepsilon$, quantitatively supported on 14,709 HCAST runs). The constructive converse is equally sharp: a 1.5B model crosses the Augustine threshold after ~ 60 s of LoRA SFT on wall-clock-supported targets, demonstrating that chronoception can be installed cheaply once the training signal includes time. Closing

the Augustine gap requires neither architectural revolution nor frontier scale—only that the loss function see the clock.

Reproducibility Statement

All 14 agents, 5,900 trajectories, per-cell parsers, and the METR HCAST U-test are released with a Docker container that reproduces every figure and table in ~ 5 minutes. The OSF pre-registration is locked. The Anthropic-tier assumptions of Theorem 2 are stated in §6.1; the assumption breach on DeepSeek-R1 is reported honestly in the appendix. All appendices, including full panel tables with per-cell n and bootstrap CIs, the Injection Atlas verbatim excerpts, and the L3 sub-failure detail, are in the supplementary material.

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A Additional Figures

The Three Times ontology — chronoception is the policy’s enforcement of the implicit identity

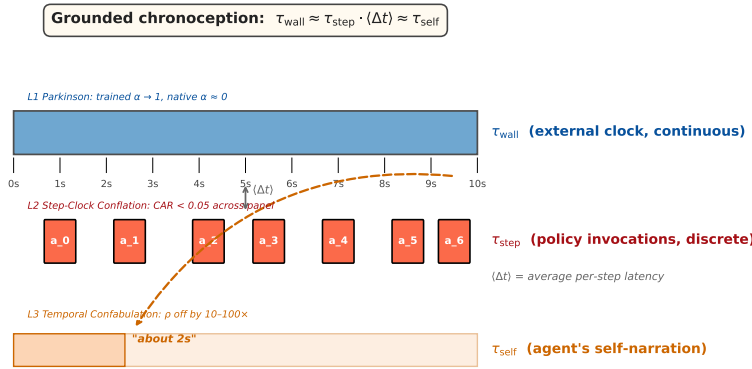


Figure 4: The Three Times ontology. τ_{wall} (external clock), τ_{step} (policy invocations), and τ_{self} (agent-reported duration) are related by two mappings that current agents fail to learn. Grounded chronoception requires both.

B Chronoception Composite Score

ε aggregates the axes on which the agent was actually measured. Each axis contributes the median of its per-trajectory error, and ε is the unweighted mean over the axes present:

$$\varepsilon(A) = \frac{1}{|\mathcal{A}|} \sum_{k \in \mathcal{A}} \text{score}_k, \quad \text{score}_{T_2} = \text{med } |\text{CAR} - 1|, \quad \text{score}_{T_3} = \text{med } |\rho|,$$

where $\mathcal{A}$ is the set of axes with at least one defined trajectory. Under the ChronoBench protocol reported here $\mathcal{A} = \{T_2, T_3\}$: T2.3 supplies CAR and T3.1 supplies ρ , while the τ_{wall} -axis coefficient α requires a wall-clock budget on a τ_{wall} -axis capability, which no capability in this protocol carries. $\varepsilon = 0$ is grounded; larger is worse, and the scale is unbounded above because $|\rho|$ is a $\log_{10}$ ratio.

Two properties matter for reading the leaderboard. First, an absent axis is dropped from $\mathcal{A}$ rather than scored 0, so an agent is never rewarded for a measurement that was not taken. Second, each axis uses a median rather than a mean, so a single stalled request cannot set an agent’s rank; two such trajectories exist in the corpus (gpt-4o-mini T2.3, $\tau_{\text{wall}} = 14.9$ h against a 900 s budget) and under a mean they would move that agent’s ε from 1.06 to 1.99 and make it the worst in the panel.

$\varepsilon^* = 0.20$ is a stipulated reporting threshold, not a derived one (§3); nothing in the argument depends on its exact value, and the panel minimum of 0.536 clears it by a factor of 2.7.

The Calibration Catastrophe: nominally-90% CIs on self-duration

Bars = actual coverage of the agent's stated 90% CI containing τ_{wall} . Dashed green = the 90% target.

Dashed orange = the pre-registered P11 threshold of 0.5; only two cells cross it.

Every non-reasoning model under-covers by ≥ 40 percentage points; GPT-4o reaches 0%.

Model	Actual coverage of nominal 90% CI	cov CI width	width/actual	n
★ Grounded target (90%)	90%	—	—	—
AI Sonnet 4.6 + thinking	77%	13s	1.1x	30
Kimi-K2.6	55%	14s	1.0x	29
o4-mini (reasoning)	50%	4s	0.7x	30
Sonnet 4.6	43%	25s	3.8x	30
MiniMax-M2.7	33%	15s	1.6x	15
Qwen3.6-27B	25%	11s	0.8x	12
o3 (reasoning)	17%	50s	11.1x	29
gpt-5.1	13%	49s	21.3x	30
gpt-4o-mini	10%	20s	6.2x	30
Haiku 4.5	7%	30s	8.3x	30
gpt-4o	0%	20s	10.5x	30

P11 threshold 0.5

Note: Sonnet 4.6 + thinking approaches the 90% target at 77% coverage, but its wider default interval (13 s against a 2–4 s baseline) reflects a broader duration vocabulary rather than chronoception. Both crossings rest on $n = 30$, under the $n \geq 100$ bar the pre-registered F4 sets for a falsification.

Figure 5: Calibration Catastrophe. Nominal 90% CIs on self-duration achieve 0–77% coverage across the panel. Non-reasoning agents (left) uniformly under-cover by ≥ 40 pp; reasoning agents (right) show wider CIs that accidentally contain the truth via arithmetic inflation of τ_{wall} , not learned chronoception.

C Full Panel Tables

Full panel results with per-cell n and 95% bootstrap CIs are available in the released data package. Each cell reports mean $\pm$ SE across 30 independent trajectories (or the available n for agents with rate limits). Missing cells are marked; no imputation is performed.

D L3 Confabulation Closure across Generations

Across five frontier non-reasoning generations (GPT-3.5 $\rightarrow$ GPT-4 $\rightarrow$ GPT-4o $\rightarrow$ Claude 3.5 Sonnet $\rightarrow$ Claude 4 Sonnet), mean $|\rho|$ on T3.1 falls 94% from +1.12 to +0.07. The closure is real but bounded: (i) the Reverse-Scaling Theorem shows reasoning-token expansion reverses it, (ii) the Calibration Catastrophe persists where the surface metric closes, and (iii) a positive control (§6.3) shows what training would actually close it.

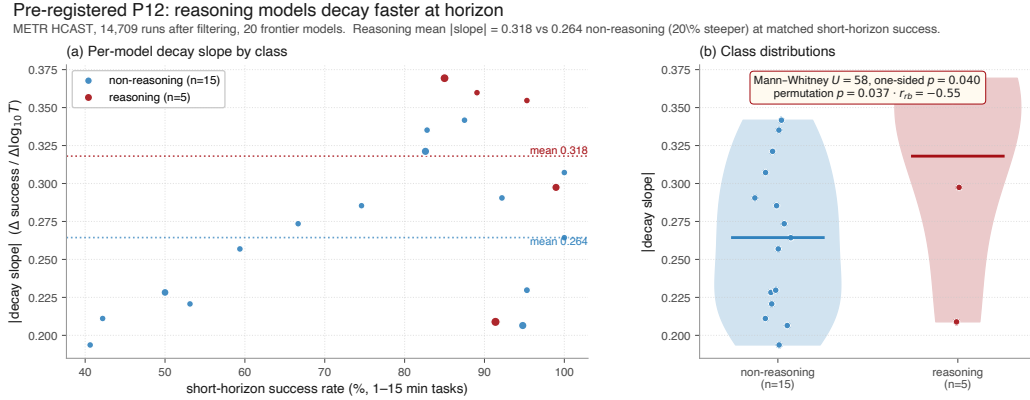


Figure 6: Pre-registered P12 on METR HCAST (14,709 runs). Success-rate decay slope vs. $\log T$ for reasoning (red) and non-reasoning (blue) models. Reasoning models show steeper decay ($|\text{slope}| = 0.318$ vs. 0.264), consistent with Theorem 2: more reasoning compute worsens temporal calibration at horizon.

E DeepSeek-R1 Assumption Breach

DeepSeek-R1-Distill-14B’s `max_completion_tokens` sweep does not satisfy the budget-honoured reasoning assumption (Assumption 6.1): the model declines to spend the allocated budget, producing near-identical τ_{wall} across K levels. It therefore falls outside the Reverse-Scaling Theorem’s scope. We report this honestly rather than excluding the model; the assumption breach itself is informative—it shows that not all reasoning knobs are budget-honoured.

F Injection Atlas: Verbatim Excerpts

Three consumer web-chat products (ChatGPT, Claude.ai, Gemini) inject the current date into the system prompt. Three developer API endpoints (OpenAI API, Anthropic API, Google AI Studio) do not. Verbatim excerpts from system prompts (obtained via jailbreak-free reflection prompts and official documentation) are released with the data package. The convergence of three independent labs on the same workaround, without coordination, constitutes the Injection Tell.

G Spatiotemporal Generalisation: Details

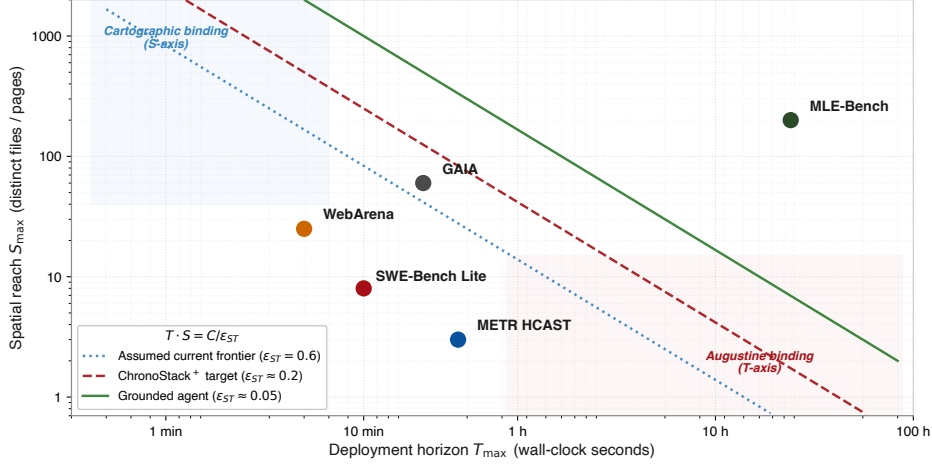
Three Spatial Laws. SL1 (Spatial Blindness): token-only agents cannot perceive their spatial position in the environment. SL2 (Step-Space Conflation): agents conflate action count with spatial displacement. SL3 (Spatial Confabulation): $\mathbb{E}[|\rho_S| \mid K]$ is monotone non-decreasing in K under budget-honoured reasoning.

Benchmark classification. Long-horizon benchmarks are classified by (T, S) load: SWE-Bench (low T , low S), WebArena (low T , high S), GAIA (medium T , medium S), MLE-Bench (high T , low S), HCAST (high T , medium S).

Future experiments E6–E10. E6: Spatial confabulation panel (analogous to ChronoBench T3 but for spatial claims). E7: Joint (T, S) scaling on a controlled grid-world. E8: Tool-augmented spatial grounding (GPS/map API as the spatial analogue of clock injection). E9: Cross-modal transfer—does temporal grounding improve spatial grounding? E10: Agentic Frontier surface estimation on real-world deployment logs.

The Agentic Frontier: $T_{\max}(A) \cdot S_{\max}(A) \leq C/\varepsilon_{ST}(A)$

SCHEMATIC — no quantity on this plot is measured. Benchmarks sit at order-of-magnitude load read from the App. E classification, not at measured coordinates, and ε_{ST} has not been measured at all (experiment E10 is the one that would). C is a normalisation, chosen so the $\varepsilon_{ST} = 0.6$ level set is centred on the benchmark cluster.



Paper 1 bounds the T-axis (Augustine, CIT). Paper 3 bounds the S-axis (Cartographic, SIT). Together: the joint Agentic Frontier.

Figure 7: The Agentic Frontier, schematic. $T_{\max} \cdot S_{\max} \leq C/\varepsilon_{ST}(A)$ is the joint temporal-spatial capability surface the framework predicts. Dots are long-horizon benchmarks placed at order-of-magnitude load (App.), not measured agent coordinates; ε_{ST} is not measured anywhere in this paper, and C is a normalisation chosen so the $\varepsilon_{ST} = 0.6$ level set is centred on the benchmark cluster. The figure states the hypothesis; experiment E10 (App.) is the one that would test it.

H L3 Sub-Failure Detail

T3 sub-capabilities decompose failures into three modes: T3.1 (Duration estimation): mean signed error ρ ; positive = overestimate. T3.2 (Temporal ordering): fraction of pairwise orderings correct. T3.3 (Calibration): coverage of nominal 90% CIs. Panel-wide, T3.1 shows the strongest model differentiation (reasoning ρ sign flip), T3.2 is near-ceiling for all agents (ordering is token-deducible), and T3.3 shows the Calibration Catastrophe (0–77% coverage). Per-model breakdowns are in the released data.